Status	Finished
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Completed	Thursday, 29 October 2026, 7:55 PM
Duration	4 mins 4 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Quick Sort
- c. Linear Search
- d. Bubble Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It simplifies the sorting process by using only one pass
- b. It allows for efficient parallel processing and sorting of data
- c. It avoids the need for recursion
- d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **3**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Duplicates
- b. Searching
- c. Selection
- d. All of the above

The correct answer is: All of the above

Question **4**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Complexity
- b. Merging
- c. Searching
- d. Sorting

The correct answer is:
Complexity

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Quick Sort
- b. Insertion Sort
- c. Bubble Sort
- d. Merge Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Merge Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It recursively sorts subproblems
- b. It bubbles up the largest element in each pass
- c. It uses the heap data structure
- d. It divides the list into sublists

The correct answer is: It bubbles up the largest element in each pass

Question **8**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It has no effect on the search operation
- c. It makes the search operation slower
- d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching

Question **9**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,80,90,150,200,300,400,470,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 200,470,80,150,40,90,300,400,70,120
- d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Linear Search
- b. Binary Search
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting is rarely used in programming
- b. Sorting makes it harder to search for items
- c. Sorting decreases the efficiency of selection operations
- d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **12**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It repeatedly finds the minimum element and moves it to the sorted part of the list
b. It compares adjacent elements and swaps them if necessary
c. It divides the list into two halves, sorts each half, and then merges them
d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **13**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes it easier to select items based on their relationship to the rest of the items
b. It makes the data unsorted
c. It slows down the process
d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **14**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is sorted independently
b. The parts are merged without sorting
c. Each part is searched for a specific element
d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **15**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Comparing each element with the others
b. Combining sorted sublists
c. Dividing the input into smaller subproblems
d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems